

END-USER USER GUIDE

PI CHESS OPEN DATABASE

PCOD 1.0 BETA “Steinitz”

Operating, research and chess-data management manual

1. File 2. Database 3. Game 4. Tools 5. Manage extensions 6. Information

1. Open Database 2. Create Database 3. Import PGN 4. Export PGN 5. Create game 6. Load engine 7. Install extensions

3. Database (3) Steinitz_Research.pgn

5. Steinitz, Wilhelm (2650) - Zukertort, Johannes (2600) 0-1

World Championship 1886 | New York | 1886.01.11 | Round 1 | C67

Symbols Comments Analyze Board image Graphics Copy FEN

Game notation

1. e4 e5 2. ♘f3 ♙c6 3. ♚f6 4. O-O ♙e4 5. ♙e1 ♙d6 6. ♙e5 ♙e7 7. ♙f1 ♙xe5 8. ♙xe5 O-O ♙d4 ♙f6 10. ♙e2 ♙e8 11. c3 ♙xe2 12. ♙xe2 ♙f5 13. ♙d3 ♙d5 14. ♙f4 ♙c6 15. ♙d2 ♙g5 16. ♙g3 ♙xg3 17. ♙xg3 ♙g7 18. ♙h5 ♙h8 19. ♙f3 ♙h6 20. ♙e1 ♙d7 21. ♙e5 ♙xe5 22. ♙xe5 ♙e8 23. ♙f4 ♙xe5 24. ♙xe5 ♙c8 25. ♙e2 ♙e6 26. ♙g4 ♙c5 27. ♙h3 ♙xd4 28. ♙xd4 ♙c1+ 29. ♙f2 ♙xb2 30. ♙d3 ♙xa2 31. ♙g3 ♙a5 32. ♙d1 ♙a4 33. ♙c2 ♙c4 34. ♙h7+ ♙f8 35. ♙h8+ ♙e7 36. ♙f6+ ♙d7 37. ♙f5 ♙c3+ 38. ♙h2 ♙xd4 39. ♙xf7+ ♙c6 40. ♙xe6+ ♙c5 41. ♙d6+ ♙c4 42. e6 ♙f4+ 43. ♙xf4+ ♙xf4 44. e7 ♙a3 45. e8= ♙a2 46. ♙a4+ ♙c3 47. ♙xa2 ♙d4 48. ♙b3+ ♙d2 49. ♙d3+ ♙e1 50. ♙xd4 0-1

Clear filters Select games 3,9-11 Example: 3,9-11 selects game 3 and games 9 through 11 Go to game Game number Search Export search results

ID	White	Black	Event	Time	Result	ECO	White Elo	Black Elo	Total moves	Source
1	Steinitz, Wilhelm	Zukertort, Johannes	World Championship 1886	1886.01.11	0-1	C67	2650	2600	99	PCOD Manual
2	Steinitz, Wilhelm	Bardeleben, Curt von	Hastings 1895	1895.08.17	1-0	C54	2600	2520	49	PCOD Manual
3	Pillsbury, Harry	Steinitz, Wilhelm	Vienna 1898	1898.06.01	1/2-1/2	D30	2570	2580	59	PCOD Manual
4	Steinitz, Wilhelm	Blackburne, Joseph	London 1872	1872.07.20	1-0	C25	2620	2550	49	
5	Lasker, Emanuel	Steinitz, Wilhelm	New York 1894	1894.03.10	1-0	C62	2700	2590	116	
6	Steinitz, Wilhelm	Zukertort, Johannes	World Championship 1886	1886.02.08	1-0	D50	2650	2600	125	

Selected game #1 | Steinitz, Wilhelm - Zukertort, Johannes | Viewing game #1 | Found 6 games

Steinitz_Research.pcod | 6 games | PCOD Hybrid 0.2

PCOD unified research interface.

Item	Value
Document version	1.0
Software version	PCOD 1.0 BETA “Steinitz”
Database format	PCOD Hybrid 0.2
Release date	17/06/2026
Owner	SpiderPawn Việt Nam
Contact	spiderpawn@hoccovua.com

Scope. This manual describes the GUI and behaviour of PCOD 1.0 BETA “Steinitz”, including Database, Clipbase, search, annotation, engine, extensions, board preview, backup and troubleshooting.

Document control

Field	Content
Document ID	PCOD-UG-1.0-BETA-STEINITZ
Status	BETA / Controlled copy
Audience	Users, coaches, researchers and data administrators
Language	English
GUI source	PCOD 1.0 BETA “Steinitz” latest build

Normative conventions

- **MUST:** mandatory requirement for correct and safe operation.
- **SHOULD:** recommended practice; may be varied with a justified reason.
- **MAY:** optional operation.
- **Database:** a PCOD data directory with the .pcod suffix, not a single file.
- **Clipbase:** a session-only virtual Database held in memory.

Structured contents

Section	Content
1	Introduction, scope and data principles
2	System requirements and portable startup
3	User-interface overview
4	Database lifecycle and Database bar
5	Clipbase workflows
6	Import, cleaning and duplicate removal
7	Creating a game and setting FEN
8	Browsing, filtering, sorting and selection
9	Advanced search and search algorithms
10	Game study and navigation
11	Move editing and variation management
12	NAGs, comments, diagrams and graphics
13	Game information and custom tags
14	Move-linked audio
15	Loading a UCI engine and analysis
16	Export, statistics and integrity checking
17	Extensions, languages and appearance
18	Display controls, board preview and shortcuts
19	Backup, recovery and performance
20	Troubleshooting and support requests
A	Default keyboard shortcuts
B	PGN tags and duplicate criteria
C	Standard operating workflows
D	Glossary and BETA limitations

Version note. The display name is 1.0 BETA “Steinitz”; release filenames use 1.0-BETA-Steinitz. Clipbase is temporary data and is not a backup mechanism.

1. Introduction, scope and data principles

Mục tiêu. Establish a common interpretation of PCOD, its data objects and safe operating principles.

PCOD (Pi Chess Open Database) is a desktop application for managing chess-game collections. It supports a reproducible research workflow: source PGN is imported into a Database, search conditions are made explicit, games are studied or annotated in one workspace, and results are exported in standard formats.

1.1. Core capabilities

- Create, open, close, arrange and merge PCOD Hybrid 0.2 Databases.
- Stream PGN imports, clean PGN, remove semantic duplicates and validate integrity.
- Search game headers, positions, material, move sequences and arbitrary PGN tags.
- Study main lines and variations; edit variations, NAGs, comments, Diagram markers and graphics.
- Use Clipbase to collect temporary games across several Databases in one session.
- Load an external UCI engine, export PGN/SVG and attach audio to moves.

1.2. Data-preservation principles

Principle	Required practice
A Database is a directory	Back up the entire .pcod directory, including index, payload, extras and sound/. Do not copy only games.pci or games.pcg.
Close before backup	Wait for active tasks, then use Choose file > Close database before copying.
Do not edit binary files	Never edit games.pci, names.pcn, games.pcg or extras with a text or hex editor.
Export before destructive work	Before large deletions, duplicate cleanup or physical deletion, export PGN or make a full copy.
Separate source and result	Keep source PGN immutable; clean to a new file; merge into a new Database.

Responsibility. The user is responsible for lawful use of PGN, audio, engines and third-party extensions. Read the EULA under Information before redistribution or use outside the permitted personal, educational and non-commercial scope.

2. System requirements and portable startup

Mục tiêu. Run PCOD without breaking the portable layout and identify the correct user-data locations.

Component	Windows portable	Linux Appliance
Operating system	Windows 10/11 x64	Linux x86_64 desktop with X11 or Wayland
Display	1366×768 minimum; 1920×1080 recommended	1366×768 minimum; 1920×1080 recommended
Memory	4 GB minimum; 8 GB+ for large collections	4 GB minimum; 8 GB+ for large collections
Java	Private runtime under runtime/; launcher	Java 21 is embedded in the Appliance

	may download it on first run	
Write access	Application and Database directories	~/.local/share/pcod and the Database directory
Engine	Windows x64 UCI executable	Executable Linux x86_64 UCI binary

2.1. Windows portable startup

Prerequisites

- The full ZIP has been extracted to a local directory.
- The application is not being run from inside the ZIP.
- Internet is available on the first run if runtime/ is empty.

Procedure

1. Open the portable directory.
2. Run PCOD.vbs or the supplied no-console launcher.
3. Wait while the private runtime is prepared if necessary.
4. Confirm that the title contains PCOD 1.0 BETA “Steinitz”.

Expected result

The main window opens without a console window and displays the menu and command bar.

Verification

- app/, lib/, runtime/, fonts/, licenses/, sbom/ and logs/ are present.
- No immediate SEVERE entry appears in logs/pcod.log.

Warning. Do not move PCOD.vbs, pcod.jar or runtime/ separately from the portable structure.

2.2. Linux Applmage startup

5. Open a terminal in the download directory.
6. Run: `chmod +x PCOD-1.0-BETA-Steinitz-Linux-x86_64.Applmage`
7. Run: `./PCOD-1.0-BETA-Steinitz-Linux-x86_64.Applmage`
8. Allow the first launch to populate the cache; subsequent launches reuse it.

Location	Purpose
~/.local/share/pcod	Application data, extensions and related settings.
~/.cache/pcod-applmage	Extracted Applmage cache.
User-selected .pcod directory	Chess Database; prefer a local drive with backup.

Reliability. Do not keep an open Database in a cloud-sync directory that locks or synchronizes individual files. Close the Database before synchronization.

3. User-interface overview

Mục tiêu. Identify each interface region so that procedures can be matched directly to the current GUI.

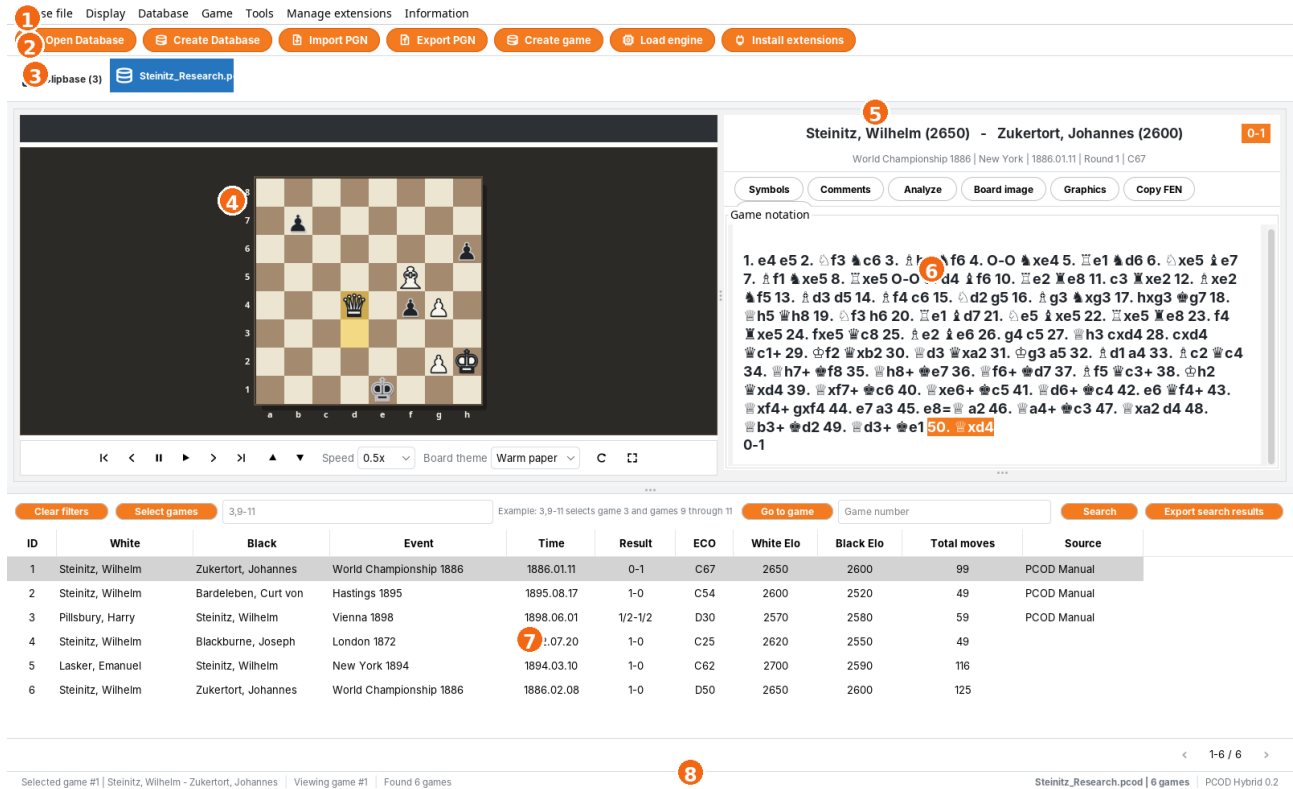


Figure 3-1. Eight principal regions of the PCOD 1.0 BETA “Steinitz” interface.

No.	Region	Purpose
1	Menu bar	Choose file, Display, Database, Game, Tools, Manage extensions and Information.
2	Command bar	Open/Create Database, Load PGN, Export PGN, Create game, Load engine and Install extensions.
3	Database bar	Clipboard and pinned/recent Database icons.
4	Board workspace	Current position, last move, graphics, navigation and board appearance.
5	Game header	Players, event, site, date, ECO, result and source status.
6	Notation/analysis	Main line, variations, comments, position data and engine output.
7	Filters and game list	Selection, go-to, search, metadata columns, paging and context commands.
8	Unified status bar	Selected/viewed game, active source, game count and Database format.

3.1. Focus and selection rules

- Single-click a row to select and load the game.
- Ctrl-click or Shift-click selects several rows for batch commands.
- Clicking a move changes the current node; arrow keys continue to navigate moves rather than moving a text caret.
- Clicking Clipboard or a Database icon changes the table source as if switching Databases.

4. Database lifecycle and Database bar

Mục tiêu. Create, open, pin, arrange, merge, remove and physically delete Databases in a controlled manner.

Operation	Action	Result
Create Database	Use Create Database and choose a .pcod directory.	PCOD creates the manifest, index, payload and extras, opens it and adds it to the Database bar.
Open Database	Use Open Database or double-click its Database-bar icon.	The list, filters, status and study workspace switch to that Database.
Close Database	Choose Choose file > Close database.	Playback and pending loads stop; the UI is cleared without exiting PCOD.

4.1. Database context menu

- **Open database:** open or switch to it.
- **Paste Clipbase:** append copies of Clipbase games; Clipbase remains unchanged.
- **Delete duplicate games:** retain the first game under the name + move tree + time rule.
- **Remove from research workspace:** remove only the icon; disk data remains.
- **Physical delete:** delete the complete Database directory after confirmation.

Mandatory distinction. Remove from research workspace does not delete data. Physical delete cannot be undone by PCOD and requires a backup.

4.2. Drag-and-drop merge

9. Select the source Database icons.
10. Hold the left mouse button and drag; a plus sign indicates copy/merge.
11. Drop to initiate creation of a new target Database.
12. Review the report for sources, imported/skipped games and copied audio.
13. Open the new Database and verify counts.

Source immutability. Merge does not remove, delete or modify the source Databases. Custom information-tag definitions are combined in source order, up to five tags.

5. Clipbase workflows

Mục tiêu. Use Clipbase as a virtual Database for temporary cross-Database collection without changing source games.

Clipbase is the first item on the Database bar. Selecting it updates the list, search, paging, metadata editing and study workspace exactly like selecting a Database.

Prerequisites

- A Database is open and one or more games are selected.

Procedure

14. Right-click the selection.
15. Choose Copy to Clipbase.
16. Observe the Clipbase count.
17. Select Clipbase and verify the games.

Expected result

Clipbase contains copies of the selected PGN; source games remain unchanged.

Verification

- Search, paging and replay work in Clipbase.

- Exact duplicate PGN copies are not added indefinitely.

5.1. Clipbase context menu

- Export Clipbase to PGN.
- Delete duplicate games using the same semantic criteria as a Database.
- Clear Clipbase.

Session-only data. Clipbase is lost when the application exits. Paste it into a Database or export it to PGN before closing PCOD.

6. Import, cleaning and duplicate removal

Mục tiêu. Import PGN with explicit checks, clean into a new file, recover safely and remove duplicates under a documented rule.

18. Open or create a Database.
19. Click Load PGN.
20. Choose the .pgn file and preserve the source file.
21. Monitor progress and do not close the application.
22. Review imported, skipped and warning counts.
23. Reset filters if the new games are not immediately visible.

Case	PCOD behaviour
Large PGN	Streaming import; the whole file is not loaded into RAM.
Comments/variants/NAGs	Compressed PGN payload preserves the full tree and annotation.
Simple main line	MOVE16 may be used for compact storage.
Invalid FEN	The PGN can be preserved by the safe codec; viewing displays a warning/fallback.
Single malformed game	Per-game rollback prevents orphan payload data.

6.1. PGN cleaning

- Use Choose file or Tools > Clean PGN.
- Choose a different output file.
- Compare counts and warnings before importing the cleaned result.
- Never overwrite the source when reproducibility matters.

6.2. Semantic duplicate removal

PCOD keeps the first occurrence. A later game is duplicate when Event, White, Black and Round match; the complete move tree including variations matches; and Date, UTCDate, UTCTime, Time and TimeControl match. Comments, Site and custom tags do not distinguish duplicates.

Verification. Record counts and export a check PGN before and after cleanup. Do not use duplicate cleanup to collapse independently annotated versions when those annotations are important.

7. Creating a game and setting FEN

Mục tiêu. Create a game from the initial position or FEN, optionally enter movetext in advance and save to a file or Database.

Save destination

☒ Save as a PGN file ☐ Append to Database

Initial position **FEN position**

Create a new game from the initial position.

Enter moves for the game

Enter SAN/PGN movetext, for example: 1. e4 e5 2. Nf3 Nc6. This field may be left blank.

Cancel **Create game**

Figure 7-1. Create game dialog with destination, initial/FEN position and movetext.

Destination	Prerequisite	Result
Save to PGN file	No Database required.	Creates a new .pgn file.
Append to Database	A Database must be open.	Adds a new game without overwriting existing games.

7.1. Movetext

- Enter SAN/PGN, for example: 1. e4 e5 2. Nf3 Nc6 3. Bb5 a6.
- PCOD validates the sequence before creation.
- Leave it blank to create only an initial/FEN position and edit later.
- Movetext must be legal from the selected FEN.



Figure 7-2. Board editor, piece palette and FEN commands.

7.2. FEN validation

24. Select the FEN position tab.
25. Paste FEN and apply it to the board.
26. Check side to move, castling rights and en-passant square.
27. Place or remove pieces directly if required.
28. Enable invalid FEN only to preserve special source data.

Invalid FEN. Preserving invalid FEN does not make the position suitable for reliable board search or engine analysis. Correct it first whenever possible.

8. Browsing, filtering, sorting and selection

Mục tiêu. Construct a working set with column filters, global sorting, paging and explicit selection expressions.

Column	Filter input	Notes
White/Black	Partial player name	Case-insensitive matching according to search implementation.
Event	Partial event text	Combined with other columns using AND.
Date	YYYY, YYYY.MM or YYYY.MM.DD	Missing components may be shown as ?.
Result	1-0, 0-1, 1/2-1/2 or *	Has a defined priority when sorted.
ECO	For example C30	Sortable ascending/descending.
Elo	Exact number	Use Advanced search for ranges.
Move count	Exact number	Displayed after conversion from stored ply count.
Custom tags	Tag value	Up to five columns after Move count.

8.1. Select games expression

- 3 selects game ID 3.
- 3,9-11 selects ID 3 and range 9 through 11.
- Whitespace is ignored; out-of-range values do not create selections.

- Go to game navigates to the page containing one specified ID.

Choose file Display Database Game Tools Manage extensions Information

Open Database Create Database Import PGN Export PGN Create game Load engine Install extensions

Clipbase (3) Steinitz_Research.p

Steinitz, Wilhelm (2650) - Zukertort, Johannes (2600) 0-1

World Championship 1886 | New York | 1886.01.11 | Round 1 | C67

Symbols Comments Analyze Board image Graphics Copy FEN

Game notation

1. e4 e5 2. ♖f3 ♗c6 3. ♜b5 ♜f6 4. O-O ♜xe4 5. ♜e1 ♜d6 6. ♜xe5 ♜e7 7. ♜f1 ♜xe5 8. ♜xe5 O-O 9. d4 ♜f6 10. ♜e2 ♜e8 11. c3 ♜xe2 12. ♜xe2 ♜f5 13. ♜d3 d5 14. ♜f4 c6 15. ♜d2 g5 16. ♜g3 ♜xg3 17. hxg3 ♜g7 18. ♜h5 ♜h8 19. ♜f3 h6 20. ♜e1 ♜d7 21. ♜e5 ♜xe5 22. ♜xe5 ♜e8 23. f4 ♜xe5 24. fxe5 ♜c8 25. ♜e2 ♜e6 26. g4 c5 27. ♜h3 cxd4 28. cxd4 ♜c1+ 29. ♜f2 ♜xb2 30. ♜d3 ♜xa2 31. ♜g3 a5 32. ♜d1 a4 33. ♜c2 ♜c4 34. ♜h7+ ♜f8 35. ♜h8+ ♜e7 36. ♜f6+ ♜d7 37. ♜f5 ♜c3+ 38. ♜h2 ♜xd4 39. ♜xf7+ ♜c6 40. ♜xe6+ ♜c5 41. ♜d6+ ♜c4 42. e6 ♜f4+ 43. ♜xf4+ gxf4 44. e7 a3 45. e8= ♜a2 46. ♜a4+ ♜c3 47. ♜xa2 d4 48. ♜b3+ ♜d2 49. ♜d3+ ♜e1 50. ♜xd4 0-1

Clear filters Select games 3,9-11 Example: 3,9-11 selects game 3 and games 9 through 11 Go to game Game number Search Export search results

ID	White	Black	Event	Time	Result	ECO	White Elo	Black Elo	Total moves	Source
1	Steinitz, Wilhelm	Zukertort, Johannes	World Championship 1886	1886.01.11	0-1	C67	2650	2600	99	PCOD Manual
2	Steinitz, Wilhelm	Hastings 1895	1895.08.17	1-0	C54	2600	2520	49	PCOD Manual	
3	Pillsbury, Harry	Vienna 1898	1898.06.01	1/2-1/2	D30	2570	2580	59	PCOD Manual	
4	Steinitz, Wilhelm	London 1872	1872.07.20	1-0	C25	2620	2550	49		
5	Lasker, Emanuel	New York 1894	1894.03.10	1-0	C62	2700	2590	116		
6	Steinitz, Wilhelm	World Championship 1886	1886.02.08	1-0	D50	2650	2600	125		

Selected game #1 | Steinitz, Wilhelm - Zukertort, Johannes | Viewing game #1 | Found 6 games

Steinitz_Research.pcod | 6 games | PCOD Hybrid 0.2

Figure 8-1. Context menu for one or more selected games.

8.2. Context commands

- Edit game information: enabled for exactly one game.
- Add information tag: applies the same name/value to all selected games.
- Copy to Clipbase.
- Show only / Hide / Delete selected games.
- Export PGN or SVG.

Reset filters. Reset restores column filters, advanced search, visibility state and original Database order.

9. Advanced search and search algorithms

Mục tiêu. Build documented queries and understand how condition groups are combined.

Advanced criteria are combined with column filters, visibility and sorting. Entered conditions are generally ANDed; individual fields may provide lists, ranges or wildcard matching.

Group	Typical criteria	Use
Game information	Players, Event, Site, Round, result, date range, ECO, Elo, move count, PGN phrases	Build a corpus by event, player or period.
Board position	FEN, exact match, pawn structure, pawns by file, variations	Opening or endgame structure research.
Material/position	Min/max queens, rooks, bishops, knights, pawns; side to move; final only	Find material configurations or endings.
Move sequence	SAN sequence; ? represents exactly one move	Find tactical or opening patterns.

Algorithm	Meaning	Example
Contains	Tag contains the entered text.	Event contains World Championship.
Exact	Complete tag value must match.	Round = 11.
Wildcard	* matches many characters; ? matches one.	White = Steinitz*.
Fuzzy	Allows small spelling differences.	Zukertort near Zukertor.

10. Game study and navigation

Mục tiêu. Read board state, main line and variations correctly and navigate without modifying data.

Figure 10-1. A game opened at the end of the main line.

Control	Function
First / Home	Initial game or FEN position.
Previous / Left	Move one node backward.
Play/Pause / Space	Auto-advance at the chosen speed; pauses audio.
Next / Right	Move one node forward in the active branch.
Last / End	Go to the end of the active branch.
Up/Down	Cycle sibling variations.
F	Flip the board.
F11	Toggle full screen.

- The status strip above the board shows side to move, last move and pre-move comment.
- The Position tab reports last move, variation count, Diagram and graphics.
- Last-move highlighting and arrows follow the active resource pack.
- Clicking notation always synchronizes the board to that node.

11. Move editing and variation management

Mục tiêu. Add or restructure the move tree without confusing main line, variation root and continuation.

Command	Availability	Effect
Promote to main line	Selected move is in a variation	Promotes the current variation at its branch

		point.
Delete following moves	Any valid node	Keeps the selected node and deletes every following move/variation.
Delete variation	Node is in a variation	Deletes the correct variation root and its subtree.
NAG/Comment/Diagram	Annotatable node	Edits annotation at the clicked node.
Audio	Move node	Attach or remove move-linked audio.

11.1. Adding a variation on the board

29. Select the branch point.
30. Choose the add-variation command where available.
31. Play the first alternative move on the board.
32. Continue the variation; PCOD follows the new node.
33. Press Esc before the first move to cancel pending variation mode.

No universal undo. Export an important game before deleting continuations, deleting variations or promoting a branch.

12. NAGs, comments, diagrams and graphics

Mục tiêu. Apply consistent annotation so that PGN remains readable, exchangeable and verifiable.



Figure 12-1. Figurine Notation, variations, NAGs and comments.

Annotation	Shortcut/gesture	Recommended use
Pre-move comment	Ctrl+Shift+A	Plan, hypothesis or context.
Post-move comment	Ctrl+A	Consequence, evaluation or lesson.
NAG panel	Shift+1 or !	Standard move/position evaluation.
Green graphic	Alt + click/drag	Primary relation or candidate.
Yellow graphic	Alt+Ctrl + click/drag	Attention or secondary candidate.
Red graphic	Alt+Shift + click/drag	Danger or tactical refutation.
Blue graphic	Alt + right-click/drag	Defensive relation.

Orange graphic	AltGr/right-click or Alt+Ctrl/right-click	Custom emphasis.
Clear all graphics	Ctrl+Alt+Y	Remove all graphic comments in the game.

Annotation convention. Use pre-move comments for plans, post-move comments for consequences, NAGs for standardized assessment and graphics for spatial relationships.

13. Game information and custom tags

Mục tiêu. Edit metadata without modifying movetext and add up to five controlled information columns.

34. Select exactly one game.
35. Right-click > Edit game information.
36. Edit Event, Site, Date, Round, White, Black, Result, Elo, ECO and displayed tags.
37. Save; PCOD updates PGN payload and indexed columns consistently.
38. Reopen and verify headers and unchanged movetext.

Movetext preservation. The information dialog does not edit moves. Use the study workspace for move-tree and annotation changes.

This Database currently has 1/5 extended information tags. The new tag will appear as a column after “Total moves”.

Tag name

Value

Figure 13-1. Adding an information tag to one or several games.

- Each Database supports at most five custom tag names.
- New columns appear immediately after Move count.
- Multi-selection applies one tag name and value to every selected game.
- Tag names start with a letter and contain only letters, digits, _, : or -.
- Tag values remain inside PGN payload; the column list is stored in information-tags.properties.

Data dictionary. Before batch tagging, define tag name, value type, units, allowed values and ownership.

14. Move-linked audio

Mục tiêu. Attach and play audio by node while preserving Database structure during backup and merge.

Property	Limit
Formats	MP3, FLAC, OGG
Maximum size	50 MB per file
Cardinality	One audio file per move
Storage	Copied into the Database sound/ directory
Playback	Can play automatically when the node is reached

Merge	Referenced files are copied to the new Database
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39. Select a move.
40. Open its context menu and choose Insert audio.
41. Choose a supported file and wait for copying.
42. Navigate away and return to test playback.
43. Use Remove audio to detach the link.

Do not delete manually. Do not remove files from sound/ while the Database is open. References and merge/backups may depend on them.

15. Loading a UCI engine and analysis

Mục tiêu. Connect an external engine, allocate resources and interpret output without treating it as authoritative game data.

Figure 15-1. UCI engine loader with depth, MultiPV, threads and hash.

Setting	Meaning	Initial recommendation
Depth	Target search depth.	10-16 for quick checks, then increase.
Lines	MultiPV alternatives.	1-3.
Threads	Engine CPU threads.	Leave capacity for PCOD and the OS.
Hash (MB)	Engine transposition-table memory.	128-512 MB depending on RAM.

44. Click Load engine.
45. Select a trusted UCI executable.
46. Set parameters appropriate to the computer.
47. Choose a position and click Analyze.
48. Read score, depth and principal variations in the engine tab.
49. Click a line move to preview; it does not change the game unless manually inserted.

Security. An engine is an external executable. Use only trusted binaries with the correct architecture and execute permission. PCOD does not bundle an engine.

16. Export, statistics and integrity checking

Mục tiêu. Export exactly the intended scope and evaluate technical health before and after major operations.

Command	Scope	Notes
Command-bar Export PGN	All current query results	Honours filters, advanced search, sorting and visibility.
Selected-game Export PGN	Selected rows only	Useful for a small corpus or pre-edit backup.
Export Clipbase to PGN	All Clipbase games	Perform before exit if not pasted into a Database.
Export SVG	Current position or selected games	Suitable for documents; support depends on the exported context.
Statistics	Open Database	Shows total records, active and deleted games.
Validate integrity	Open Database	Checks CRC, payload pointers and related structures.

50. Record the displayed result count.
51. Export to a new PGN file.
52. Import the export into a test Database or another PGN tool.
53. Compare count, headers, main lines, variations and comments for a sample.
54. Store a checksum when the export is used for research or handover.

Integrity failure. If validation reports an error, stop all writes, close the Database, make an unchanged backup and provide logs/pcod.log with a sanitized copy.

17. Extensions, languages and appearance

Mục tiêu. Install and manage code-free extensions in three categories without granting Database code execution.

Manage PCOD language, interface and search-algorithm extensions in one place.

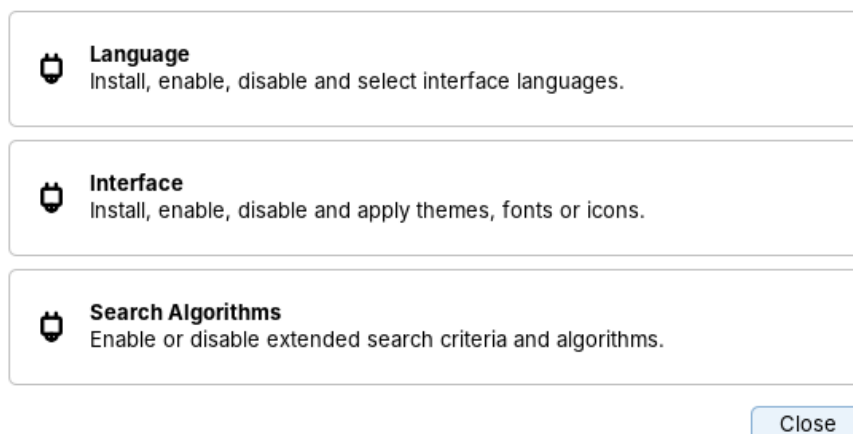


Figure 17-1. Central Manage extensions dialog.

Category	Format/capability	Controls
Language	.pcod-language; UTF-8 messages, BCP 47, LTR/RTL	Install, enable/disable, select, English fallback.
Interface	.pcod-resource; board themes, PNG pieces, fonts, icons	Install, preview, enable/disable, select, safe fallback.
Search algorithms	Data-only criteria/algorithms	Enable/disable Contains, Exact, Wildcard and Fuzzy.

The screenshot displays the PCOD 1.0 BETA "Steinitz" interface. At the top, a menu bar includes "Choose file", "Display", "Database", "Game", "Tools", "Manage extensions", and "Information". Below this, a toolbar contains buttons for "Open Database", "Create Database", "Open extension manager...", "Create game", "Load engine", and "Install extensions". The "Manage extensions" menu is open, showing options: "Language", "Interface", "Search Algorithms", and "Open extension manager...".

The main window is divided into two panes. The left pane shows a chessboard with a game in progress. The right pane displays game information for "Steinitz, Wilhelm (2650) - Zukertort, Johannes (2600)" from the "World Championship 1886 | New York | 1886.01.11 | Round 1 | C67". It includes tabs for "Symbols", "Comments", "Analyze", "Board image", "Graphics", and "Copy FEN". The game notation is shown, with the 50th move highlighted: "50. xxd4".

Below the main panes, there is a table of games. The table has columns: ID, White, Black, Event, Time, Result, ECO, White Elo, Black Elo, Total moves, and Source. The table lists 6 games, with the first game being "Steinitz, Wilhelm vs Zukertort, Johannes" from the "World Championship 1886".

ID	White	Black	Event	Time	Result	ECO	White Elo	Black Elo	Total moves	Source
1	Steinitz, Wilhelm	Zukertort, Johannes	World Championship 1886	1886.01.11	0-1	C67	2650	2600	99	PCOD Manual
2	Steinitz, Wilhelm	Bardeleben, Curt von	Hastings 1895	1895.08.17	1-0	C54	2600	2520	49	PCOD Manual
3	Pillsbury, Harry	Steinitz, Wilhelm	Vienna 1898	1898.06.01	1/2-1/2	D30	2570	2580	59	PCOD Manual
4	Steinitz, Wilhelm	Blackburne, Joseph	London 1872	1872.07.20	1-0	C25	2620	2550	49	
5	Lasker, Emanuel	Steinitz, Wilhelm	New York 1894	1894.03.10	1-0	C62	2700	2590	116	
6	Steinitz, Wilhelm	Zukertort, Johannes	World Championship 1886	1886.02.08	1-0	D50	2650	2600	125	

At the bottom of the interface, there is a status bar showing "Selected game #1 | Steinitz, Wilhelm - Zukertort, Johannes | Viewing game #1 | Found 6 games" and "Steinitz_Research.pcod | 6 games | PCOD Hybrid 0.2".

Figure 17-2. Manage extensions menu and categories.

55. Open Manage extensions from the menu or extension icon.
56. Select the appropriate category.
57. Review package author, version and licence.
58. Install and enable it if required.
59. Select the language/resource/algorithm and verify the UI.
60. Disable or remove a faulty package; PCOD falls back safely.

Security model. Resource Pack 1.0 and Language Extension 1.0 are data-only and cannot access a Database or execute code. Install only verifiable packages.

18. Display controls, board preview and shortcuts

Mục tiêu. Adapt the workspace, use hover preview when study space is hidden and configure non-conflicting shortcuts.

Choose file Display Database Game Tools Manage extensions Information

Open Database Import PGN Export PGN Create game Load engine Install extensions

Clipboard (3)

Clear filters Select games 3,9-11 Example: 3,9-11 selects game 3 and games 9 through 11 Go to game Game number Search Export search results

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Selected game #1 | Steinitz, Wilhelm - Zukertort, Johannes | Viewing game #1 | Found 6 games

Steinitz_Research.pcod | 6 games | PCOD Hybrid 0.2

Figure 18-1. Display menu for bars, workspace, command buttons and language.

- Show/hide the full command bar.
- Show/hide the Database bar.
- Show/hide the complete research workspace independently.
- Show/hide individual command buttons.
- Change the interface language.

Choose file Display Database Game Tools Manage extensions Information

Open Database Import PGN Export PGN Create game Load engine Install extensions

Clipboard (3)

Clear filters Select games 3,9-11 Example: 3,9-11 selects game 3 and games 9 through 11 Go to game Game number Search Export search results

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5	Lasker, Emanuel	Steinitz, Wilhelm	New York 1894	1894.03.10	1-0	C62	2700	2590	116	
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Selected game #1 | Steinitz, Wilhelm - Zukertort, Johannes | Viewing game #1 | Found 6 games

Steinitz_Research.pcod | 6 games | PCOD Hybrid 0.2

Figure 18-2. Hovering a game shows the final main-line position when the research workspace is hidden.

- Preview is active only when the research workspace is hidden.
- Hover a row; after a short delay the final main-line position appears.
- The popup uses the same board component, theme, pieces, orientation and last-move marking.
- It closes on mouse exit, scrolling, page/source change or when the workspace is shown.

- It works for both Database and Clipbase.

Nhập tổ hợp theo dạng Ctrl+N, Ctrl+Shift+N hoặc Delete. Để trống để vô hiệu hóa phím tắt. Không được dùng trùng một tổ hợp cho nhiều lệnh.

Tạo Database	Ctrl+N	Mặc định: Ctrl+N
Mở Database	Ctrl+O	Mặc định: Ctrl+O
Đóng Database	Ctrl+W	Mặc định: Ctrl+W
Nạp PGN	Ctrl+I	Mặc định: Ctrl+I
Xuất ván đã chọn	Ctrl+E	Mặc định: Ctrl+E
Tạo ván cờ	Ctrl+Shift+N	Mặc định: Ctrl+Shift+N
Nạp engine	Ctrl+L	Mặc định: Ctrl+L
Xóa ván đã chọn	Delete	Mặc định: Delete
Thoát phần mềm	Ctrl+Q	Mặc định: Ctrl+Q
Nước đi trước	Left	Mặc định: Left
Nước đi tiếp theo	Right	Mặc định: Right
Chuyển biến phía trên	Up	Mặc định: Up
Chuyển biến phía dưới	Down	Mặc định: Down
Về đầu ván	Home	Mặc định: Home
Đi đến cuối ván	End	Mặc định: End
Phát/Tạm dừng tự động	Space	Mặc định: Space
Lật bàn cờ	F	Mặc định: F
Bật/tắt toàn màn hình	F11	Mặc định: F11

Figure 18-3. Shortcut editor layout; labels follow the selected UI language.

61. Open Tools > Keyboard shortcuts > Customize keyboard shortcuts.
62. Enter Ctrl+N, Ctrl+Shift+N, Delete, etc.; leave blank to disable.
63. Ensure that no two actions use the same combination.
64. Save.
65. Test in the game table, notation, study workspace and full screen.

Restore defaults. Tools > Keyboard shortcuts > Default keyboard shortcuts removes custom mappings and restores all defaults.

19. Backup, recovery and performance

Mục tiêu. Protect unique data and operate large Databases with predictable performance.

66. Wait for import, export, merge, annotation save, audio copy and engine tasks to finish.
67. Close the Database in PCOD.
68. Copy the entire .pcod directory to an independent location.
69. Use an ISO-date suffix, e.g. SteinitzResearch_2026-06-17.pcod.
70. Periodically open and validate a backup.
71. Keep at least one offline or versioned copy.

Symptom	Adjustment
Search is slow	Narrow column filters first; avoid fuzzy search across the full Database unless required.
Game load is slow	Reduce concurrent work; use local SSD storage; avoid network folders.
Engine uses all CPU	Reduce threads/hash/depth or stop analysis during import/export.
Too many columns	Use only necessary custom tags; resize/temporarily hide

	the study workspace.
Many preview loads	Hold the pointer steady; preview debounces and cancels stale requests.

Automatic recovery. PCOD records import/update recovery state for rollback after interruption. After a power failure, reopen and validate integrity before further edits.

20. Troubleshooting and support requests

Mục tiêu. Diagnose with evidence, avoid changing the affected data and provide a reproducible support package.

Symptom	Checks in order
Application does not open	Re-extract; check runtime/lib; run from a writable directory; inspect logs/pcod.log.
Database does not open	Select the .pcod directory containing manifest.properties, not games.pci alone.
Game is not visible	Reset filters; confirm active source is Database or Clipbase; check hidden state.
No hover preview	Hide the research workspace; hold the pointer on a row; do not scroll.
Clipbase disappeared after exit	This is by design; export or paste before exit.
Engine fails	Check UCI binary, permission, architecture, parameters and log.
Audio fails	Check MP3/FLAC/OGG, size ≤50 MB, sound/ and audio libraries.
Integrity error	Stop writes, close, back up unchanged, do not repair binary files manually.

20.1. Minimum support package

- Exact version: PCOD 1.0 BETA “Steinitz”.
- Operating system, architecture, resolution and scaling.
- Numbered reproduction steps, expected and actual results.
- Full-window and error-dialog screenshots.
- logs/pcod.log.
- A sample PGN or sanitized Database copy.
- Time and last operation before the failure.

Report example. “Open Database A > select game #125 > click move 18... > Delete following moves. Expected..., actual.... Reproduced 3/3 times.”

A. Default keyboard shortcuts

Mục tiêu. Quick reference; every application shortcut can be changed under Tools > Keyboard shortcuts.

Action	Default
Create Database	Ctrl+N
Open Database	Ctrl+O

Close Database	Ctrl+W
Load PGN	Ctrl+I
Export selected	Ctrl+E
Create game	Ctrl+Shift+N
Load engine	Ctrl+L
Delete selected games	Delete
Exit	Ctrl+Q
Previous/next move	Left / Right
Previous/next variation	Up / Down
First/last position	Home / End
Play/pause	Space
Flip board	F
Full screen	F11
Pre/post-move comment	Ctrl+Shift+A / Ctrl+A
NAG panel	Shift+1 or !
Clear all graphics	Ctrl+Alt+Y

B. PGN tags and duplicate criteria

Mục tiêu. Standardize metadata used for import, editing, search and duplicate cleanup.

Tag	Meaning	Example
Event	Event name	World Championship 1886
Site	Location	New York
Date	YYYY.MM.DD	1886.01.11
Round	Round identifier	11
White/Black	Player names	Steinitz, Wilhelm
Result	Game result	1-0
WhiteElo/BlackElo	Ratings	2650
ECO	Opening code	C67
FEN	Non-standard initial position	r1bqk...
SetUp	Signals FEN setup	1
UTCDate/UTCTime/Time/TimeControl	Additional time fields	2026.06.17 / 12:30:00 / 600+5

Duplicate rule. Names = Event + White + Black + Round; moves = full tree including variations; time = Date + UTCDate + UTCTime + Time + TimeControl. Site, comments and custom tags do not distinguish duplicates.

C. Standard operating workflows

Mục tiêu. Provide concise SOPs for consistent work by individual users or research teams.

C.1. Build a corpus from source PGN

72. Create a new target Database.
73. Store a checksum and immutable source PGN.
74. Clean into a new file if required.
75. Import and retain the import report.
76. Remove duplicates only after exporting a check copy.
77. Apply controlled Source/Project tags.
78. Validate integrity.
79. Close and back up the Database.

C.2. Collect from several Databases

80. Open Database A; search and copy games to Clipbase.
81. Switch to Database B and continue.
82. Select Clipbase, inspect and remove duplicates if needed.
83. Export Clipbase PGN as a check copy.
84. Paste Clipbase into the target Database.
85. Close and back up the target.

C.3. Study one game

86. Select the game and verify headers.
87. Go to the main-line end and review with Left/Right.
88. Use pre-move comments for plans and post-move comments for evaluation.
89. Apply NAGs and graphics according to team conventions.
90. Create alternatives and verify main-line promotion.
91. Use engine analysis critically rather than copying automatically.
92. Export the game and test reopening.

D. Glossary and BETA limitations

Mục tiêu. Clarify important terms and operational limitations.

Term	Definition
Main line	Preferred continuous move sequence in PGN.
Variation	Alternative branch beginning at a replay node.
Node	One move/position state in the replay tree.
NAG	Numeric Annotation Glyph for standardized evaluation.
FEN	Forsyth-Edwards Notation describing a position.
UCI	Universal Chess Interface between GUI and engine.
Clipbase	Session-only virtual Database.
Logical delete	Game is marked deleted while its record may remain in statistics.
Physical delete	Complete Database directory is removed from disk.

- BETA release: keep backups and validate before relying on unique data.
- Clipbase is not persistent across application exits.
- Maximum five custom information tags per Database.
- No chess engine is bundled.

- Hover preview appears only when the research workspace is hidden.
- There is no universal undo for all destructive move-tree operations.

End of manual. Core rule: identify the correct source → define a reproducible query → operate and verify → export → close the Database → back up the full directory.